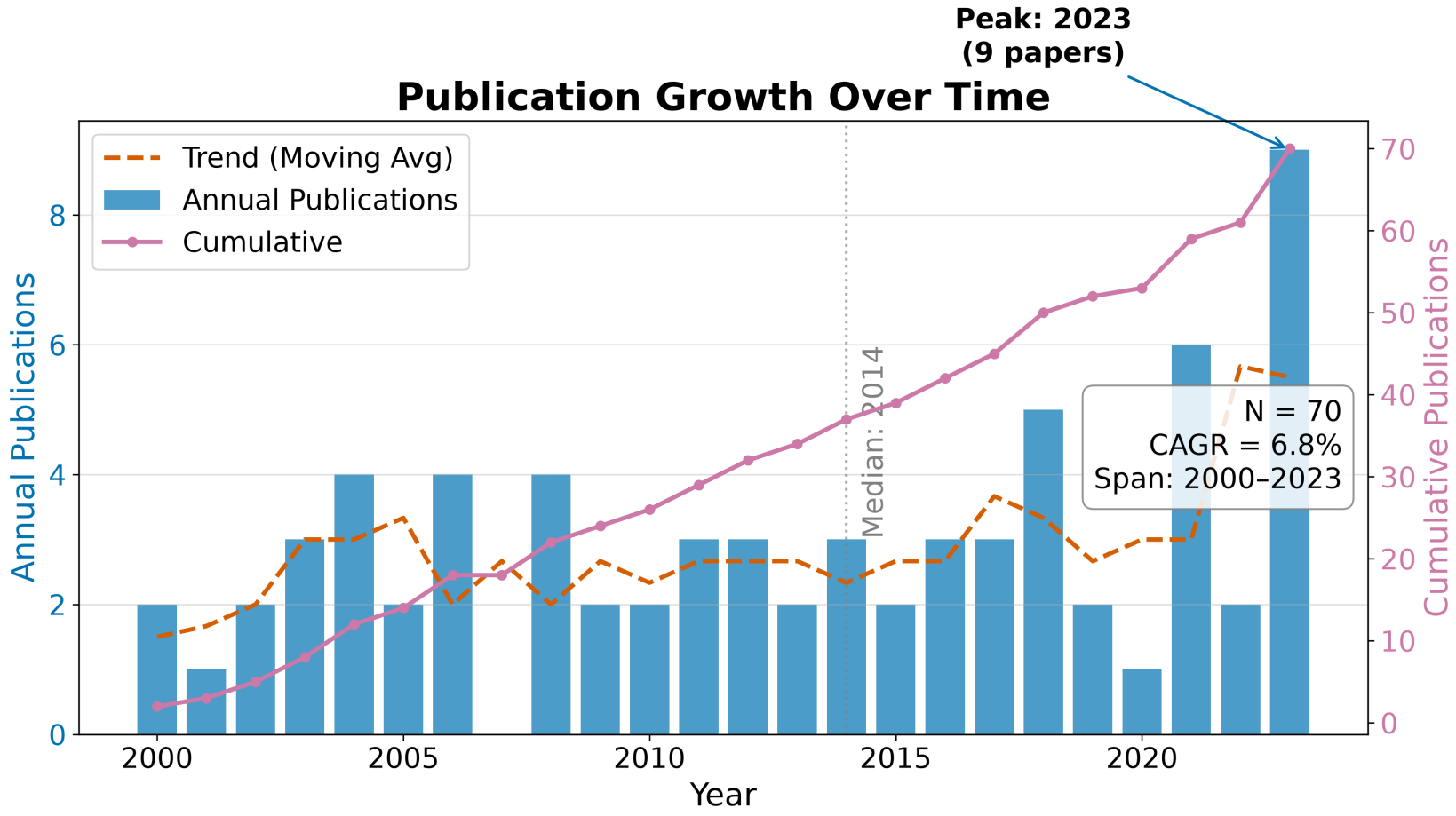
# Results: Field Overview

## Results: Field Overview I

The bibliometric analysis corpus for **Exoplanet Atmospheres** contains $N = 70$ records. Its dated temporal subset contains 70 records spanning 2000–2023 (23 years). Publication volume grows at a compound annual rate of 6.76% (mean year-over-year growth 38.9%, doubling time 2.1 years), peaking in 2023 with 9 records that year. The growth curve is a first-order descriptive summary of this analyzed corpus; it is not a field-wide estimate of research activity. The retrieved fixture contains 80 records, while the configured phase union contains 46 unique records. The figures and descriptive bibliometrics in this section use the separate 70-record analyzed corpus, with growth computed on its dated temporal subset; identifier and full-text coverage below use the retrieved fixture.

# Results: Field Overview I



## Results: Field Overview I

Figure 1: Publication growth curve for Exoplanet Atmospheres. Annual publication counts (bars) and cumulative total (line) show sustained growth from 2000 through 2023, peaking in 2023.

## RQ1: Field Size and Growth I

The temporal analysis describes a retained literature slice spanning 23 years. The compound annual growth rate of 6.76% implies a corpus doubling time of approximately 2.1 years under the configured retrieval and indexing conditions. The peak year 2023 contains 9 retained records; that count may reflect both publication activity and delays or differences in source indexing, so it should not be interpreted as a causal trend without a live coverage audit.

**Table 1. Top publication years.**

# RQ1: Field Size and Growth I

| Year | Publications |
|---|---|
| 2003 | 3 |
| 2004 | 4 |
| 2006 | 4 |
| 2008 | 4 |
| 2011 | 3 |
| 2012 | 3 |
| 2014 | 3 |
| 2018 | 5 |

# RQ1: Field Size and Growth I

| Year | Publications |
|---|---|
| 2021 | 6 |
| 2023 | 9 |

Records distribute across the 4 configured subfields as shown in Table 2, with **Observational Methods** the largest bucket at 51.4% of the classified corpus. The dominance of Observational Methods is a property of the configured taxonomy and retained corpus. It is not a domain prevalence estimate without a live, source-backed review and a documented coverage assessment.

**Table 2. Subfield distribution.**

# RQ2: Subfield Composition I

| Subfield | Papers | Share |
|---|---|---|
| Observational Methods | 36 | 51.4% |
| Atmospheric Molecules | 13 | 18.6% |
| Jwst Instruments | 11 | 15.7% |
| Theoretical Modeling | 10 | 14.3% |

# RQ2: Subfield Composition I

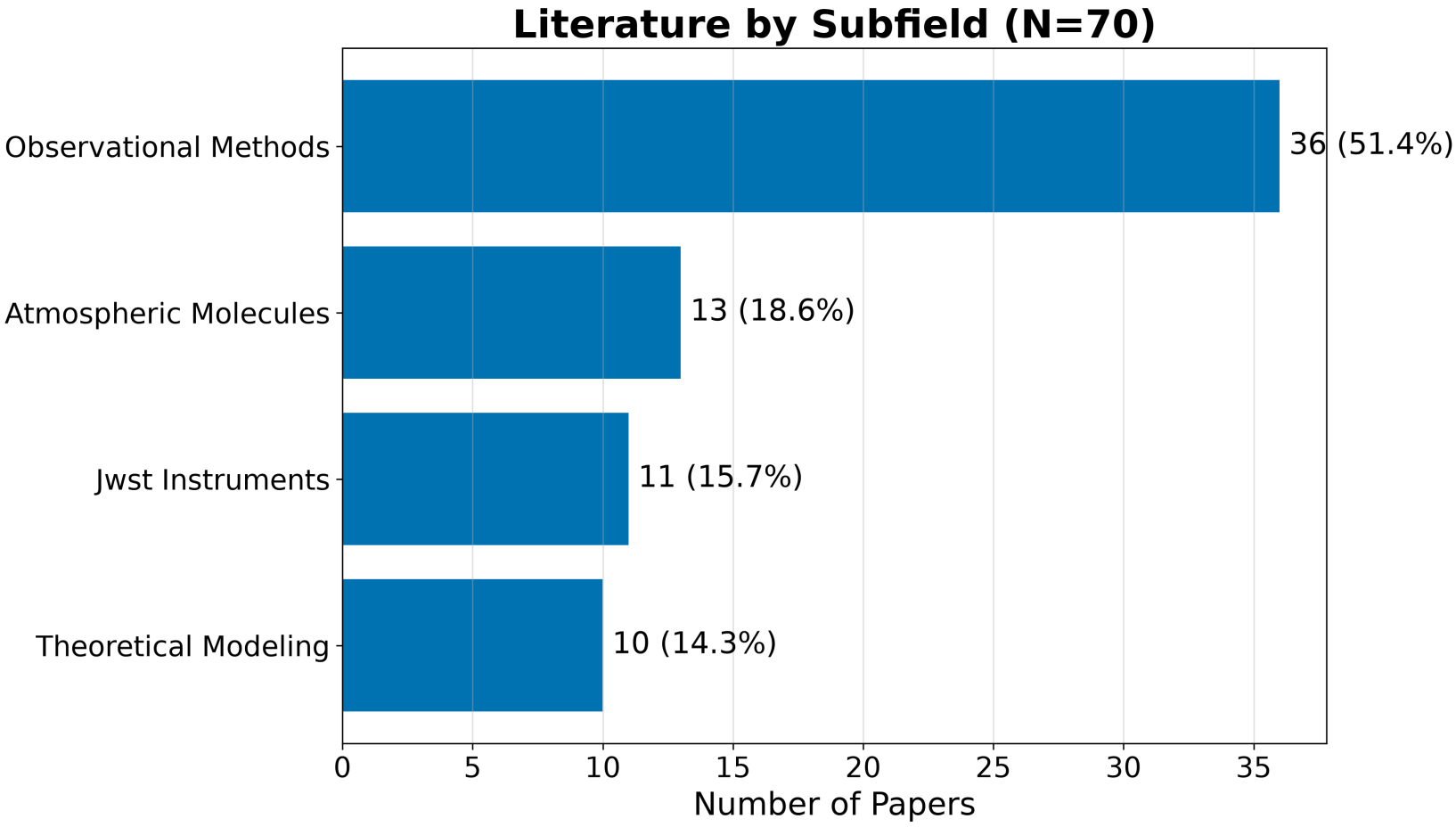


Figure 2: Field summary dashboard for Exoplanet Atmospheres. The dashboard combines corpus size, temporal range, subfield distribution, and key bibliometric indicators in a single overview panel.

# RQ2: Subfield Composition I

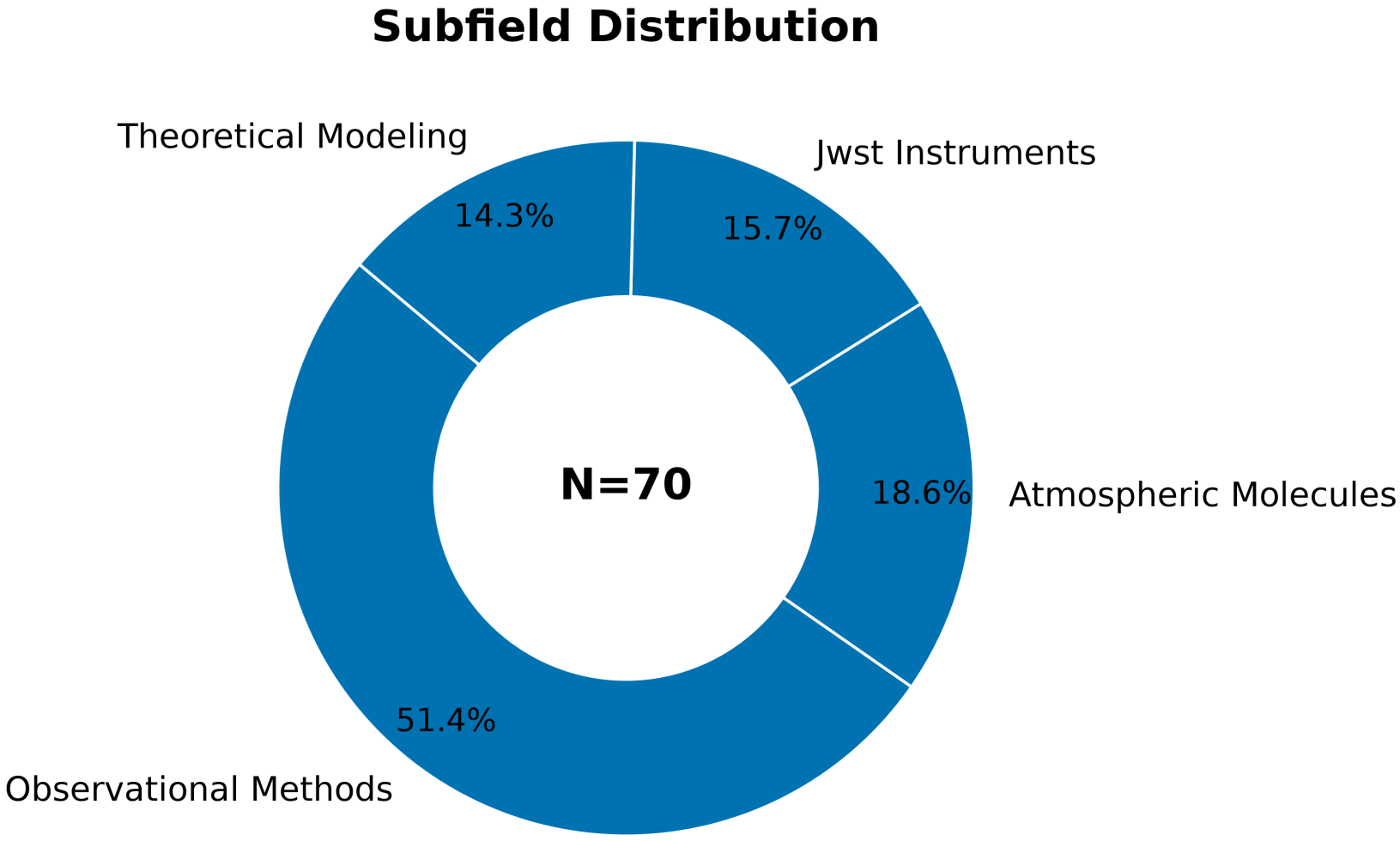


Figure 3: Subfield distribution for Exoplanet Atmospheres. The 4-bucket taxonomy shows the relative weight of each configured sub-area, with Observational Methods dominant at 51.4%.

# RQ2: Subfield Composition I

# RQ2: Subfield Composition I

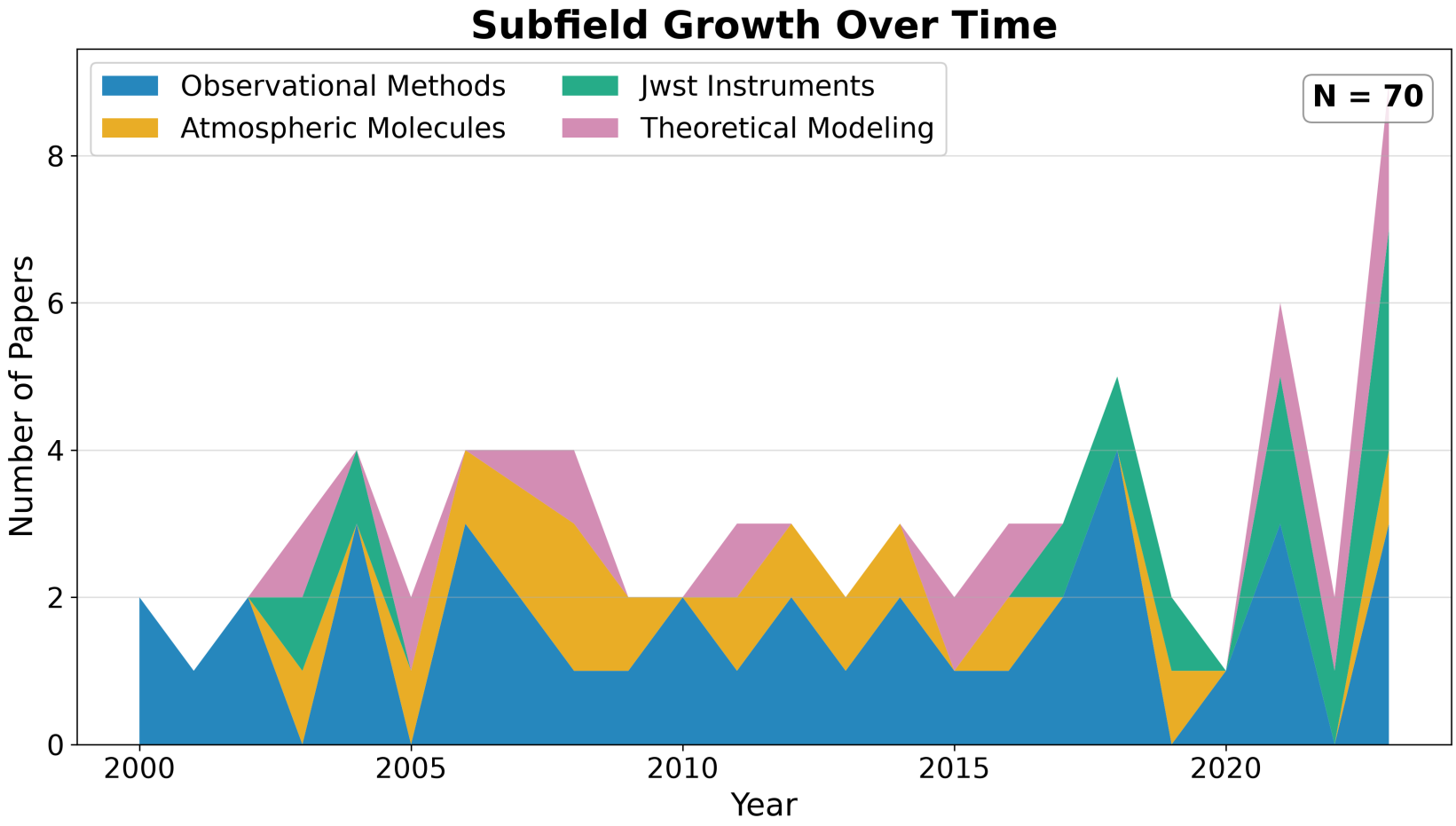


Figure 4: Subfield timeline for Exoplanet Atmospheres. Stacked annual publication counts by subfield show how each sub-area has evolved over time, revealing emerging and declining research threads.

## Identifier and Full-Text Coverage I

The retrieved fixture has measurable identifier coverage: 80 of 80 records carry DOIs, supporting cross-engine de-duplication. OpenAlex IDs are present for 12 records. Abstract coverage stands at 100.0% (80 records), which limits the text analytics to that subset. Open-access status is confirmed for 42.5% of records, and 42.5% have a direct PDF link.

The analyzed corpus spans 144 unique authors across 70 papers, yielding a mean of 1.51 papers per author. Citation counts range from zero to 110 (mean 37.0, median 31.0), with a total of 2,593 citations across the corpus. The Gini coefficient of citation concentration is 0.421. This statistic describes concentration within the retained corpus and should not be generalized to citation behavior in the field.

**Table 3. Citation count distribution.**

# Descriptive Bibliometrics I

| Citations | Papers |
|---|---|
| 0 | 0 |
| 1-9 | 12 |
| 10-49 | 39 |
| 50-99 | 16 |
| 100-499 | 3 |
| 500+ | 0 |

# Descriptive Bibliometrics I

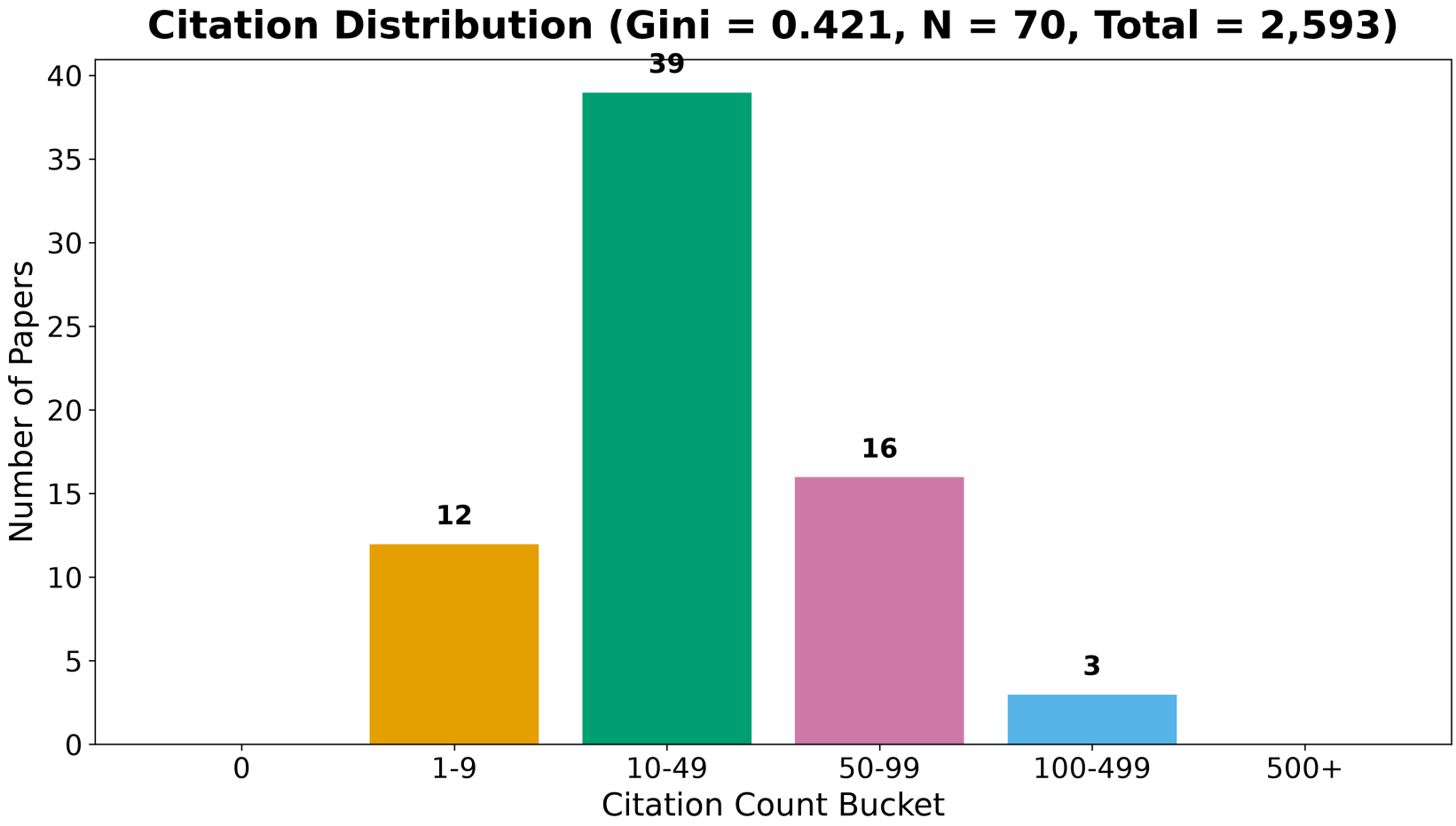


Figure 5: Citation distribution for Exoplanet Atmospheres. The histogram shows the number of papers in each citation-count bucket, with the Gini coefficient annotated. The distribution is a descriptive property of the retained corpus.

# Descriptive Bibliometrics I

**Table 4. Top publication venues.**

# Descriptive Bibliometrics I

| Venue | Papers |
|---|---|
| Astronomy and Astrophysics | 11 |
| Publications of the Astronomical Society of the Pa | 11 |
| The Astrophysical Journal | 10 |
| Monthly Notices of the Royal Astronomical Society | 9 |
| Nature Astronomy | 9 |
| Research Notes of the AAS | 9 |
| Icarus | 6 |
| The Astronomical Journal | 5 |

# Descriptive Bibliometrics I

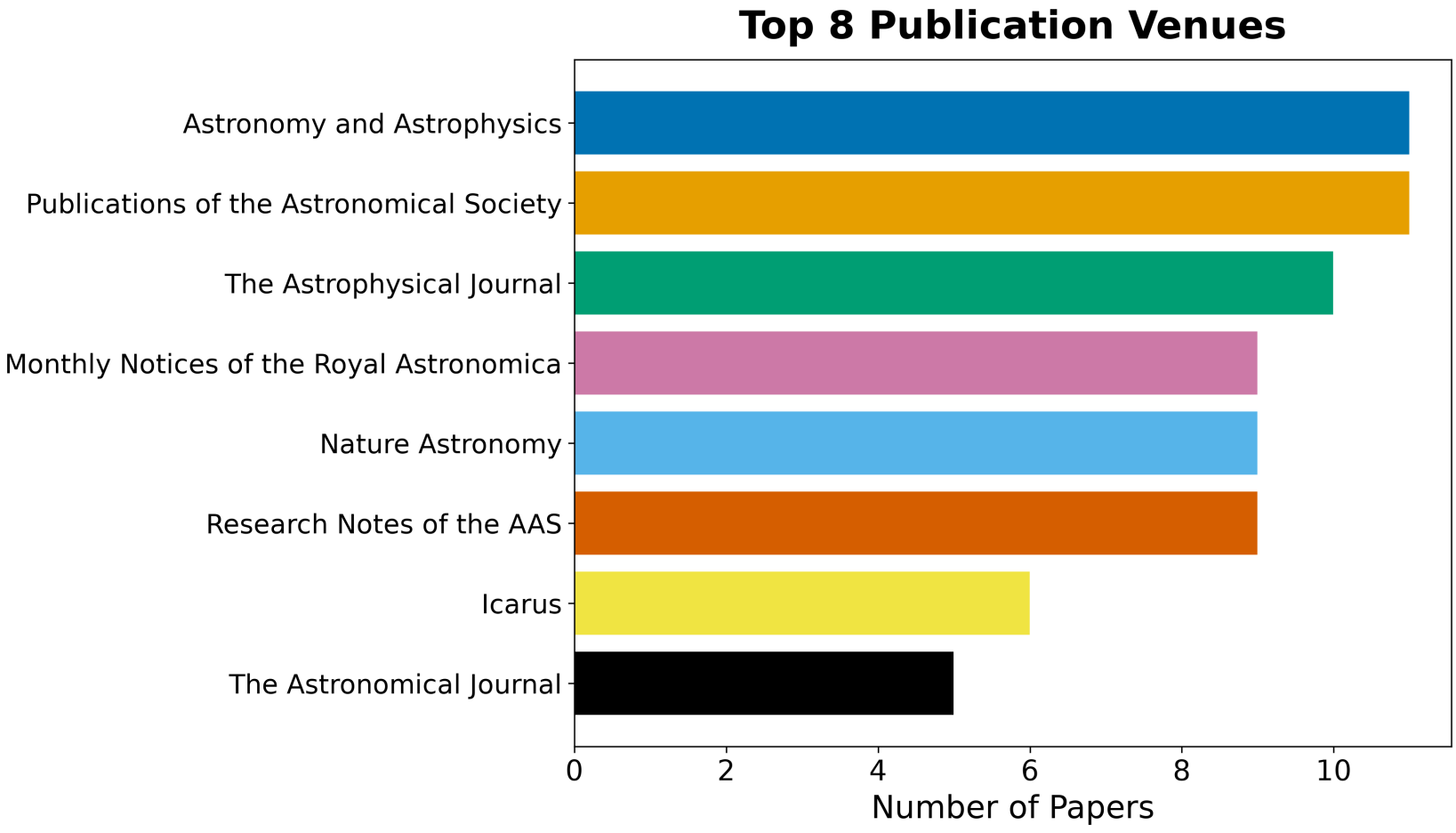


Figure 6: Top publication venues for Exoplanet Atmospheres. The horizontal bar chart shows the venues with the most retained records; it describes this corpus rather than the complete field.

**Table 5. Top authors by publication count.**

# Descriptive Bibliometrics I

| Rank | Author | Papers |
|---|---|---|
| 1 | A. Fournier | 4 |
| 2 | B. Owens | 4 |
| 3 | D. Ito | 4 |
| 4 | D. Nakamura | 4 |
| 5 | G. Ito | 4 |
| 6 | K. Ito | 4 |
| 7 | B. Kowalski | 3 |
| 8 | C. Singh | 3 |

# Descriptive Bibliometrics I

| Rank | Author | Papers |
|---|---|---|
| 9 | E. Tanaka | 3 |
| 10 | F. Lindgren | 3 |

# Descriptive Bibliometrics I

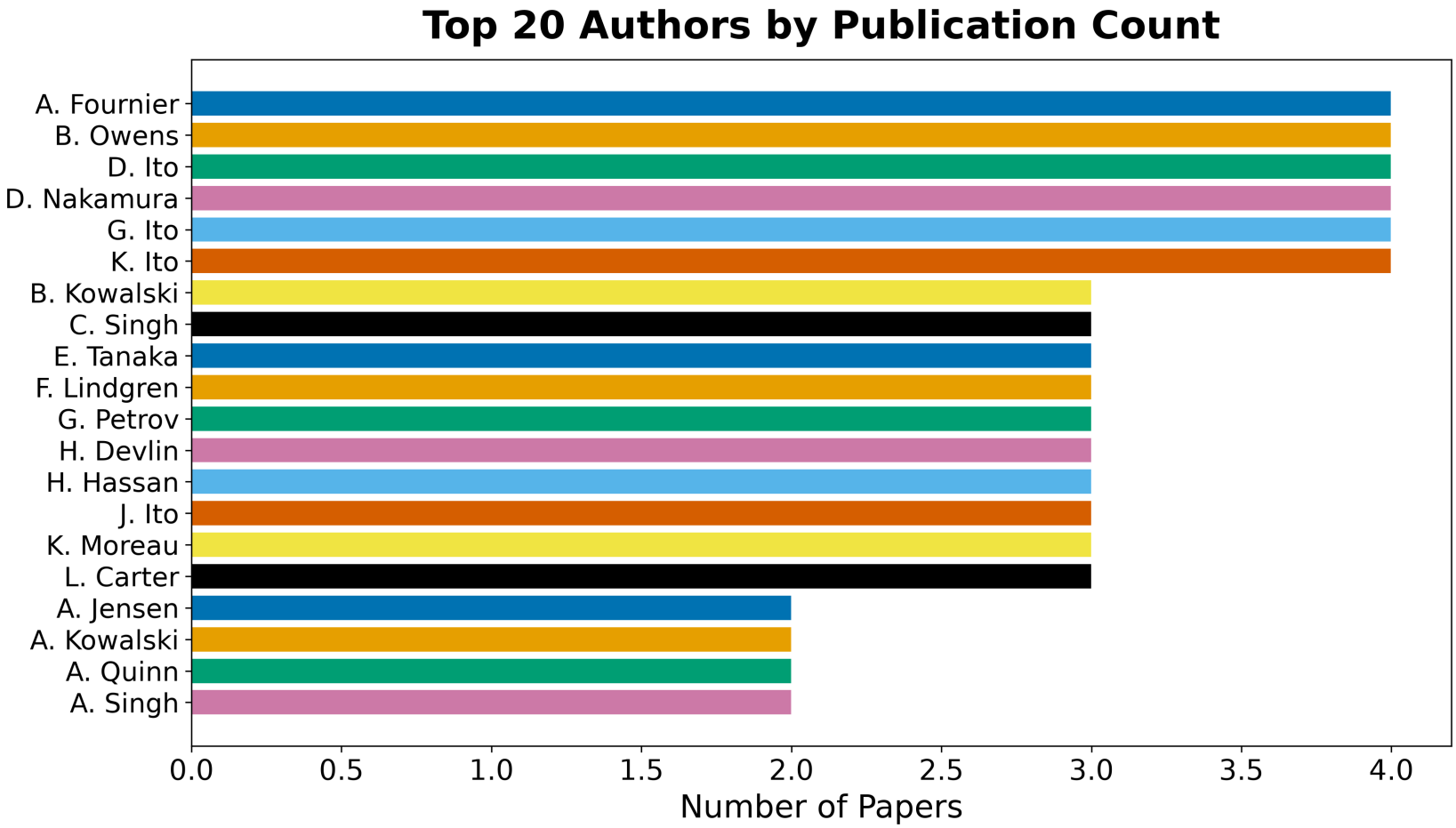


Figure 7: Author productivity for Exoplanet Atmospheres. The horizontal bar chart shows authors with the most retained records; names and counts depend on source coverage and deduplication.